

HDOC DAY – 22

QUESTION:

Write a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and use a switch statement to perform the operation selected by the user:

1. Print the multiplication tables of m and n side by side, using multipliers from 1 to 10.
2. Display a multiplication grid containing the products of integers from 1 to m and integers from 1 to n .
3. For each integer from m to n , print that integer followed by the next five consecutive integers.
4. Exit the program.

Continue displaying the menu until the user selects the Exit option. Display an appropriate error message if the user enters an invalid menu choice.

>Algorithm:

1. Start.
2. Input two positive integers m and n .
3. Check whether $m \leq n$.
4. Display the menu:
 1. Multiplication tables of m and n
 2. Multiplication grid
 3. Consecutive integers
 4. Exit
5. Read the user's choice.
6. Use a switch statement to perform the selected operation.
7. If choice is 1, print multiplication tables of m and n from 1 to 10 side by side.
8. If choice is 2, print the multiplication grid from 1 to m and 1 to n .
9. If choice is 3, for each integer from m to n , print the integer and the next five consecutive integers.
10. If choice is 4, exit the program.
11. For any other choice, display "Invalid choice".
12. Repeat steps 4–11 until choice 4 is selected.
13. Stop.

Test Case	m	n	Choice	Expected Result
1	3	5	1	Display tables of 3 and 5 side by side
2	3	5	2	Display multiplication grid from 1 to 3 and 1 to 5
3	3	5	3	Display next five consecutive integers for 3, 4 and 5
4	3	5	4	Exit the program
5	3	5	6	Display "Invalid choice"

>Code C:

```

1#include <stdio.h>
2
3int main()
4{
5    int m, n, choice, i, j;
6
7    printf("Enter m and n: ");
8    scanf("%d %d", &m, &n);
9
10   do
11   {
12       printf("\nMenu\n");
13       printf("1. Multiplication tables of m and n\n");
14       printf("2. Multiplication grid\n");
15       printf("3. Consecutive integers\n");
16       printf("4. Exit\n");
17
18       printf("Enter your choice: ");
19       scanf("%d", &choice);
20
21       switch(choice)
22       {
23           case 1:
24               for(i = 1; i <= 10; i++)
25               {
26                   printf("%d x %d = %d\t", m, i, m * i);
27                   printf("%d x %d = %d\n", n, i, n * i);
28               }
29               break;
30
31           case 2:
32               for(i = 1; i <= m; i++)
33               {
34                   for(j = 1; j <= n; j++)
35

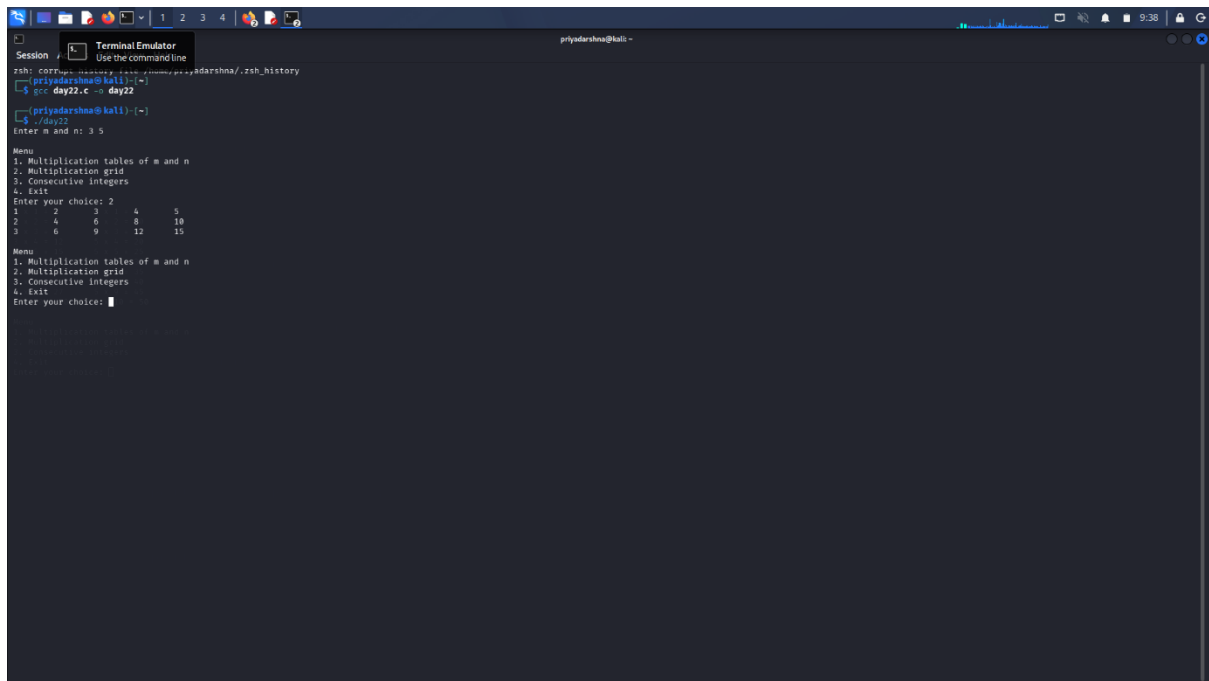
```

```
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day22.c
34     for(j = 1; j <= n; j++)
35     {
36         printf("%d\t", i * j);
37     }
38     printf("\n");
39 }
40 break;
41
42 case 3:
43     for(i = m; i <= n; i++)
44     {
45         printf("%d: ", i);
46         for(j = i + 1; j <= i + 5; j++)
47         {
48             printf("%d ", j);
49         }
50         printf("\n");
51     }
52     break;
53
54 case 4:
55     printf("Exiting program...\n");
56     break;
57
58 default:
59     printf("Invalid choice\n");
60 }
61 while(choice != 4);
62 return 0;
63 }
64 } while(choice != 4);
65 return 0;
66 }
67 }
68 }
```

>TEST CASE 1:

```
Session / Terminal Emulator
Use the command line
zsh: coredump: ./day22
[priyadarshna@kali]~$ gcc day22.c -o day22
[priyadarshna@kali]~$ ./day22
Enter m and n: 3 5
Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1
3 x 1 = 3    5 x 1 = 5
3 x 2 = 6    5 x 2 = 10
3 x 3 = 9    5 x 3 = 15
3 x 4 = 12    5 x 4 = 20
3 x 5 = 15    5 x 5 = 25
3 x 6 = 18    5 x 6 = 30
3 x 7 = 21    5 x 7 = 35
3 x 8 = 24    5 x 8 = 40
3 x 9 = 27    5 x 9 = 45
3 x 10 = 30   5 x 10 = 50
Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2
Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3
Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
Exiting program...
```

>TEST CASE 2:



A terminal window titled "Terminal Emulator" showing a C program execution. The user enters 'gcc day22.c -o day22' and runs './day22'. The program prompts for 'm' and 'n' (3 and 5). It displays a menu with options: 1. Multiplication tables of m and n, 2. Multiplication grid, 3. Consecutive integers, 4. Exit. The user selects option 2. The program then displays a 3x5 multiplication grid.

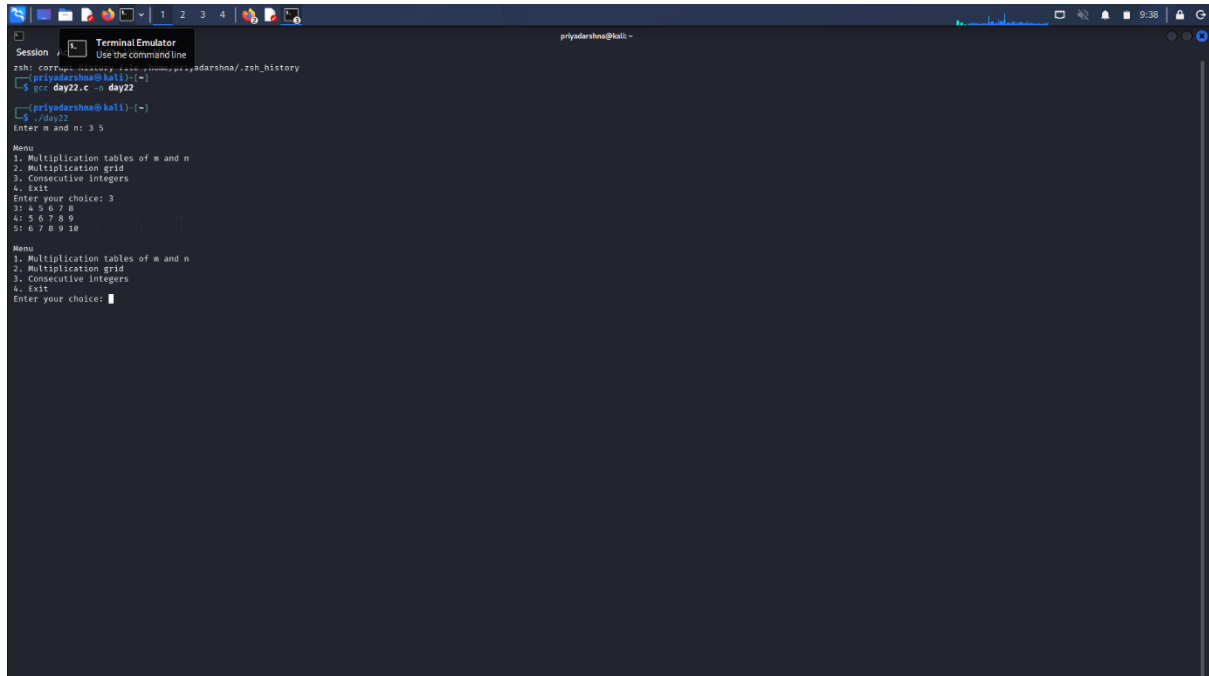
```
zsh: correct window title to priyadarshna/.zsh_history
(priyadarshna@kali)~$ gcc day22.c -o day22
(priyadarshna@kali)~$ ./day22
Enter m and n: 3 5

Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

1 2 3 4 5
2 4 6 8 10
3 6 9 12 15

Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 
```

>TEST CASE 3:



A terminal window titled "Terminal Emulator" showing the same C program execution as Test Case 2. The user enters 'gcc day22.c -o day22' and runs './day22'. The program prompts for 'm' and 'n' (3 and 5). It displays the same menu. The user selects option 3. The program then displays three lines of consecutive integers: 3 4 5 6 7 8, 4 5 6 7 8 9, and 5 6 7 8 9 10.

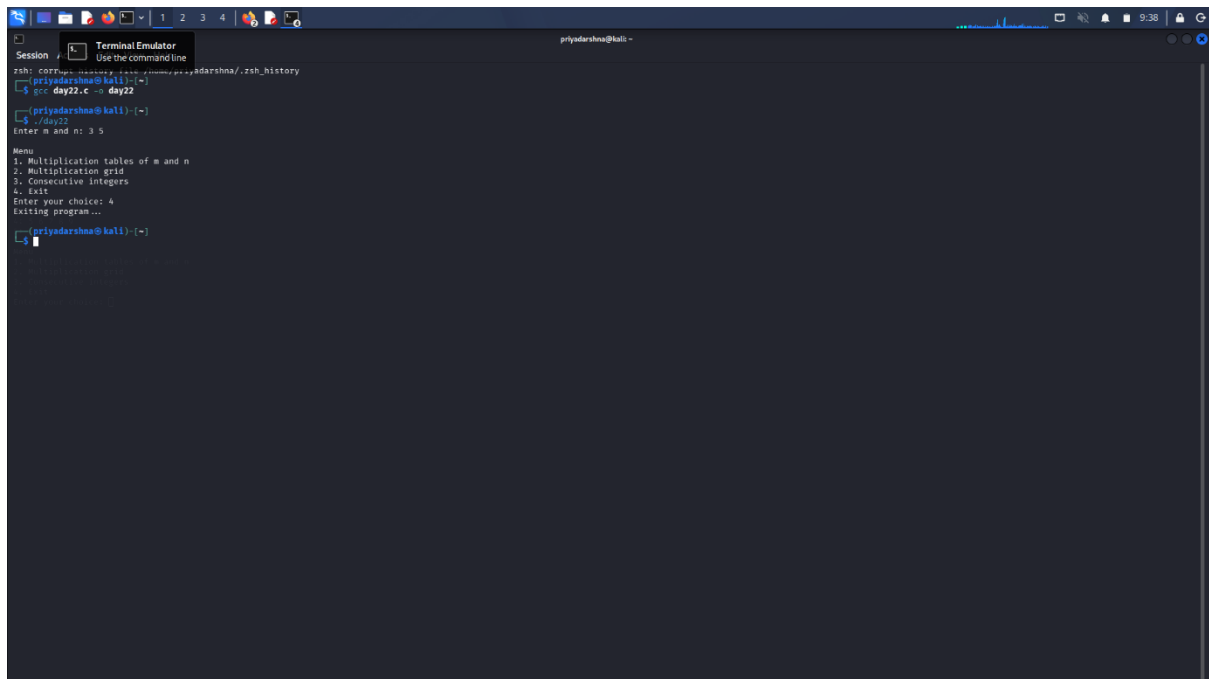
```
zsh: correct window title to priyadarshna/.zsh_history
(priyadarshna@kali)~$ gcc day22.c -o day22
(priyadarshna@kali)~$ ./day22
Enter m and n: 3 5

Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

3 4 5 6 7 8
4 5 6 7 8 9
5 6 7 8 9 10

Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 
```

>TEST CASE 4:

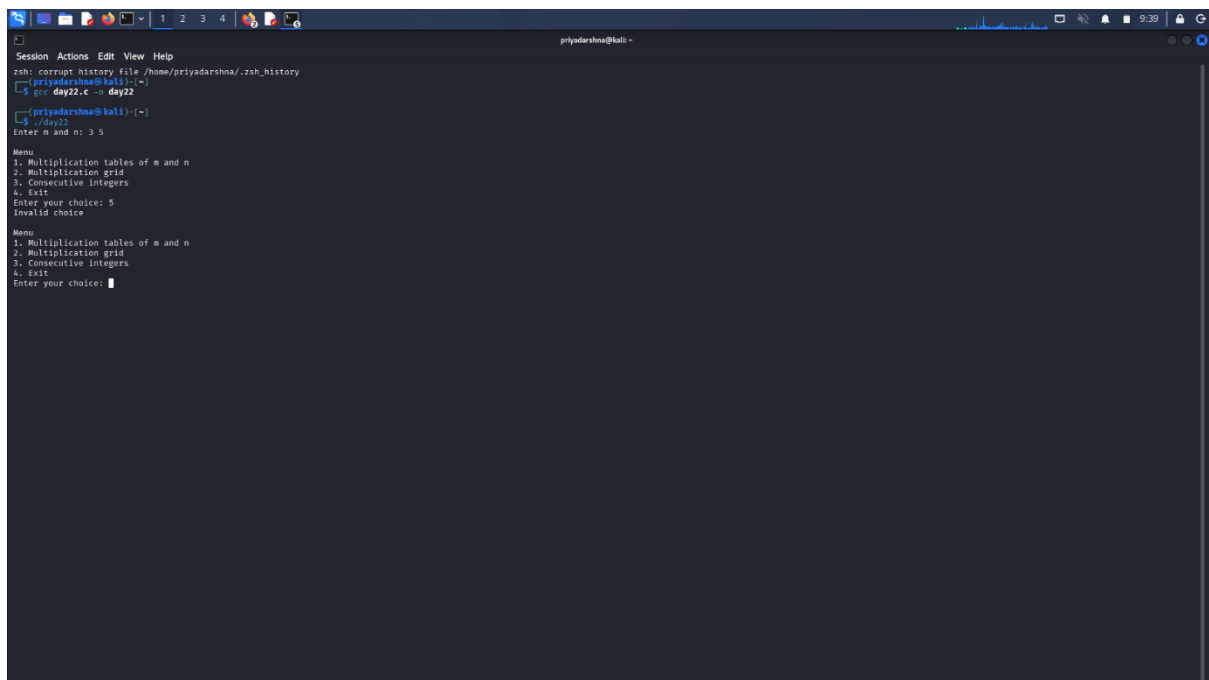


A terminal emulator window titled "Terminal Emulator" with a subtitle "Use the command line". The session shows the user running a C program named "day22.c". The program prompts for "Enter m and n: 3 5". It then displays a menu with four options: 1. Multiplication tables of m and n, 2. Multiplication grid, 3. Consecutive integers, and 4. Exit. The user enters "4", and the program outputs "Exiting program..." and returns to the shell prompt.

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day22.c -o day22
[priyadarshna@kali:~]$ ./day22
Enter m and n: 3 5

Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
Exiting program...
[priyadarshna@kali:~]$
```

>TEST CASE 5:



A terminal emulator window titled "Terminal Emulator" with a subtitle "Use the command line". The session shows the user running the same C program "day22.c". The program prompts for "Enter m and n: 3 5". It displays the same menu as in Test Case 4. The user enters "5", and the program outputs "Invalid choice". It then displays the menu again, and the user enters a space character, which is also treated as an invalid choice.

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day22.c -o day22
[priyadarshna@kali:~]$ ./day22
Enter m and n: 3 5

Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 5
Invalid choice

Menu
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 
```